Securiti Fast Track Internship Program 2026

Technical Report

For

PII Detection and Masking

Version 1.0

Submitted By:

Abdul Muizz

Submitted to:

Mr. Hamza Kazi

Date: May 17th, 2026

Data Science Internship Task

PII Detection and Masking Technical Report

Evaluation Date: 2026-05-17

In-domain Test Set: 3,650 sentences OOD

Test Set: WikiANN English, 3,652 sentences

Evaluation: seqeval strict IOB2

GitHub Repository: https://github.com/a-muizz28/pii_masking.git

1. Project Overview:

The project builds a local PII detection and masking pipeline for PERSON names and EMAIL addresses in English Wikipedia-style text. WikiNeural contained no EMAIL annotations, so the training and test splits were augmented with synthetic email addresses injected next to existing PERSON spans. Train and test email domains were separated to reduce lexical memorisation.
The final report uses three fixed approaches:

Approach A Model: DistilBERT-s42

- **Role:** Lightweight encoder baseline and best OOD encoder in this run

Approach B Model: DeBERTa-s7

- **Role:** Best in-domain DeBERTa seed

Approach C Model: LLaMA

- **Role:** Prompted local LLM baseline

2. Methodology:

- **Approach A:** DistilBERT-s42. distilbert-base-cased was fine-tuned once with seed 42. It is treated as Approach A because it had the best WikiANN PER F1 among the cached OOD runs.

- **Approach B:** DeBERTa-s7. microsoft/deberta-v3-small was trained across seeds 0, 7, and 42. Seed 7 is the selected DeBERTa checkpoint for the final A/B/C comparison.

- **Approach C:** LLaMA. LLaMA 3.2-1B-Instruct ran locally through llama-cpp-python with a JSON extraction prompt. The parser normalised extracted names and emails back to IOB2 tags for shared metric calculation.

All approaches were evaluated with strict span F1, token level false positive rate, token-level false-negative rate, and redaction leak rate. A span is counted as leaked when any token in a gold PII span is predicted as O.

3. In-Domain Results:

- **Approach A (DistilBERT-s42):** PER F1 = 0.982 | EMAIL F1 = 0.999 | Macro F1 = 0.990 | FPR = 0.0012 | FNR = 0.0073 | Leak Rate = 2.3%

- **Approach B (DeBERTa-s7):** PER F1 = 0.981 | EMAIL F1 = 1.000 | Macro F1 = 0.990 | FPR = 0.0015 | FNR = 0.0071 | Leak Rate = 2.2%

- **Approach C (LLaMA zero-shot):** PER F1 = 0.419 | EMAIL F1 = 0.426 | Macro F1 = 0.423 | FPR = 0.1653 | FNR = 0.0760 | Leak Rate = 13.0%

Paired bootstrap results (from results/day5/bootstrap_results.json):

- **DeBERTa-s7 vs LLaMA:** Observed F1 Difference = 0.417 | 95% CI = [0.406, 0.429] | One-sided p-value = 0.000

- **DistilBERT-s42 vs LLaMA:** Observed F1 Difference = 0.416 | 95% CI = [0.405, 0.429] | One-sided p-value = 0.000

4. Error Analysis:

(Day 6 error buckets stored in results/day6/error_buckets.json)

- **Approach A (DistilBERT-s42): FN**_PER = 94 | FN_EMAIL = 4 | FP_PER = 95 | FP_EMAIL = 2 | Total Errors = 195

- **Approach B (DeBERTa-s7):** FN_PER = 84 | FN_EMAIL = 0 | FP_PER = 116 | FP_EMAIL = 0 | Total Errors = 200

- **Approach C (LLaMA):** FN_PER = 1,374 | FN_EMAIL = 707 | FP_PER = 9,261 | FP_EMAIL = 3,289 | Total Errors = 14,631

Approach A and B have similar total error counts in-domain. Approach B misses fewer PERSON and EMAIL spans, while Approach A produces fewer PERSON false positives. Approach C has a much larger error volume, dominated by false-positive PERSON spans and false-positive

EMAIL spans.

5. WikiANN Cross-Domain Results

WikiANN contains no EMAIL spans in this setup, so overall_f1 is set to PER F1 for the OOD comparison.

- **Approach A (DistilBERT-s42):** PER F1 = 0.777 | FPR = 0.021 | FNR = 0.215 | Leak Rate = 19.3%
- **Approach B (DeBERTa-s7):** PER F1 = 0.718 | FPR = 0.016 | FNR = 0.274 | Leak Rate = 28.0%
- **Approach C (LLaMA):** PER F1 = 0.628 | FPR = 0.144 | FNR = 0.126 | Leak Rate = 13.7%

Approach A is the best OOD encoder by PER F1. Approach C has the lowest WikiANN leak rate because it over-predicts PERSON spans, but this comes with a token false-positive rate nearly seven times higher than Approach A.

6. Notable Findings

Approach A is the preferred default from this run: it has the strongest WikiANN PER F1 while preserving encoder-level false-positive rates. Approach B is still competitive in-domain, but its WikiANN FNR rises to 0.274 and its OOD leak rate reaches 28.0%. Approach C generalises recall better on WikiANN, but its high FPR makes it unsuitable as the primary production masker without additional precision controls. In-domain performance alone is not sufficient for release decisions. The WikiANN evaluation exposes a clear distribution shift for both encoders.

7. Production Recommendations

Use Approach A as the default model for the current project milestone. Keep Approach B as a secondary encoder benchmark, especially for tracking DeBERTa seed stability. Do not deploy Approach C as the primary masker without a precision filter or downstream review step. Add target-domain validation before any production deployment, as OOD leak rates are materially higher than in-domain leak rates. Expand training data with broader name-origin coverage to reduce cross-domain PERSON false negatives.

8. Security Considerations

A non-zero leak rate means real PII can survive masking. Approach A leaks 2.3% of in-domain PII spans and 19.3% of WikiANN PERSON spans. Those rates are too high for unsupervised release of sensitive text. The masking pipeline should be treated as a risk-reduction layer, not anonymisation under GDPR Article 4(1). Additional controls are required for quasi-identifiers, co-reference chains, rare names, and deployment domains that differ from the training data.

10. Conclusion

This project evaluated three approaches to local PII detection and masking for PERSON names and EMAIL addresses. The results demonstrate a clear performance hierarchy between fine-tuned encoders and zero-shot LLM prompting.

Approach A (DistilBERT-s42) emerges as the recommended model for deployment. It achieves a macro F1 of 0.990 in-domain, the best WikiANN PER F1 of 0.777 among all tested approaches and maintains encoder-level false-positive rates throughout. Approach B (DeBERTa-s7) is equally strong in-domain but suffers a steeper cross-domain degradation, with an OOD leak rate of 28.0% that limits its suitability as a general-purpose masker.

Approach C (LLaMA zero-shot) confirms that a small instruction-tuned LLM without fine-tuning is not viable as a primary PII masker. Despite showing relatively better recall generalisation on WikiANN, its in-domain macro F1 of 0.423, leak rate of 13.0%, and false-positive rate over 100 times higher than the encoders make it unsuitable for unsupervised production use.

A key takeaway from this evaluation is that strong in-domain performance does not guarantee production safety. Both encoders showed meaningful degradation on WikiANN, underscoring the need for target domain validation before any deployment. No approach tested here meets the bar for anonymisation under GDPR Article 4(1), and the pipeline should be treated as a risk-reduction layer requiring additional safeguards for sensitive data.

Future work should focus on broadening training name origin coverage, applying domain adaptive pre-training, and establishing held out evaluation on the intended production domain prior to release.